SoK: Transfer Boundaries of LLM and Agent Side-Channel Defenses

***Abstract*—When does a side-channel defense result remain valid after an agent system changes? We systematize defense transfer boundaries: the emitter, executed state, calibration population, attacker observation and evaluation environment. A map of 82 study families, 39 source-bound result records and a nine-item audit distinguish supported transfers from incompatible comparisons. Controlled demonstrations expose the consequences: upstream padding does not protect a new emitter; read-cover reasoning does not establish state preservation for writes; calibration and attacker refitting change measured protection; and timing reveals signals hidden by length-only evaluation. A two-endpoint transfer matrix shows that moving calibration can exchange protection between targets: cancellation balanced accuracy rises from 50.0% to 70.8% while return falls from 67.3% to 50.0%. These known-cohort measurements turn model sensitivity into a directional transfer test. Our contribution is an operational account of where defense evidence stops transferring, which measurement resolves each boundary, and how to select defenses without borrowing protection, utility or cost from a different operating point.**

1. Introduction

Encryption conceals message content, yet LLM engines and agents expose packet sizes, cache timing, visited domains and tool sequences [1], [2], [3], [4], [5], [6]. A defense result applies to a particular receiver, execution boundary and workload. Reusing a headline score across a frontend, dispatcher or tool provider can change both its protected signal and its cost object.

Our unit is the *evaluated result record*. Separating Dangol's stored/live results prevents subtracting unmatched scores; separating Early Bird's KV-prefix and semantic-cache mechanisms prevents borrowing overhead. Observer access, threat polarity and execution state explain why read-only privacy cover and state-changing evasion have different utility consequences.

Central question and contributions. *Which changes invalidate a measured defense result?* The map locates systems, records bind operating points, and the audit identifies transfer obligations.

- **Systematic transfer evidence.** The complete 39-record census separates six trace transformations, five simulation/estimate results, six component/calibration measurements, 20 executed collections/prototypes and two unresolved placements. A six-axis, 82-family matrix distinguishes reported holdouts from condition sweeps and unresolved evidence.
- **Target-dependent calibration transfer.** Moving MiniMax calibration to GLM changes cancellation/return BA from 50.0/67.3% to 70.8/50.0%. Protection shifts in opposite directions; a single aggregate transfer score hides this operating-point change.
- **Localization and utility boundaries.** Write-only shaping costs 1.45–1.51× bytes but leaves return BA at 61.3–61.6%; detector precision differs sharply across endpoints. Fresh deadline-controlled planners succeed on 37/50 versus 33/50 tasks with prefix cover. These measurements determine what oracle localization and logical state equivalence actually license.

Exp 1's DPI premise and Exp 2's isolated-state invariant are mechanism checks. The empirical contribution is the measured operating-point, target and utility behavior beyond those invariants; Table 1 links it to the SoK findings.

Scope of claims. We contribute record-level systematization and controlled demonstrations of evidence transfer. The real-agent target is a requested operation category; the known retail cohort contains 53 customers. Measurements establish fitted-suite inference, logical state conformance and scoped resource costs. A separate 100-episode live study measures prefix-shadow utility under injected deadlines. Deployment-wide security and full joint utility require the validation specified in §9.3.

Figure 1 places trace and action paths on the same agent–tool chain. This joint view makes the receiving representation and the execution boundary explicit before selecting a defense.

2. Scope and Systematization Method

2.1. Research questions, eligibility and threat polarity

We organize the systematization around four questions: which hidden targets, threat polarities and auxiliary observables appear across LLM and agent systems (RQ1); how signal generation origin relates to receiver observation permissions (RQ2); which holdouts establish portability across tasks, configurations and hardware (RQ3); and which evidence binds workflow protection, logical state preservation and resource cost, distinguishing these from full end-to-end planner utility (RQ4).

TABLE 1. RESEARCH QUESTIONS, SYSTEMATIZATION FINDINGS AND EVIDENCE. EXPERIMENTAL FOLLOW-UPS TEST THE COMPARISON RULES; THE SOK CONTRIBUTION IS THE RECORD-LEVEL SYNTHESIS.

Question	Finding	Evidence in the main paper
RQ1: targets and observables	A common signal can serve confidentiality, attribution, evasion or authorized diagnostics; polarity changes the protected interest.	Complete core-family map (§4, Table 6).
RQ2: origin versus access	B-size cover does not bind A-call order; plaintext re-emission creates a fresh B trace; shared A-call projections do not establish unchanged state. These distinctions identify the evidence required to reuse a result.	NetEcho/SplitDecoy, WRA/AHB and dispatcher read/write records (§5); Exps 1–2.
RQ3: portability	The 82-family six-axis matrix separates explicit holdouts from condition variation and unresolved evidence; input separation does not certify a changed generator.	Complete matrix (Appendix D); synthesis in §7; directional calibration matrix in Exp 3.
RQ4: protection, state and cost	Bind inference, utility and cost to one scoped operating point. The 20-family audit gives R8 Bound 3/17 for component-or-task utility and 2/16 under a task-only sensitivity.	Twenty-family R1–R9 matrix; Exps 3–4 leakage/state/cost contrasts (§8). The same audit locates the utility boundary of our demonstrations.

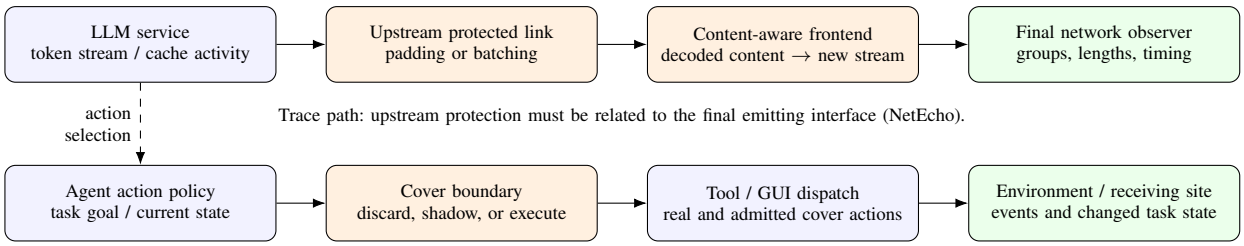


Figure 1. Observation and execution boundaries. The four demonstrations test emitter placement, stateful execution/link composition, real-agent operation signatures, and timing with matched transport controls.

Core inclusion requires five conjunctive criteria: (C1) LLM or agent execution, (C2) a hidden target property, (C3) an execution-dependent auxiliary observable rather than ordinary output semantics alone, (C4) specified observer permissions, and (C5) an empirical inference evaluation. Mixed-feature papers qualify only if they isolate an auxiliary-only C3 component; thus coding fingerprints, WildAuth, ZeroPrint, endpoint-content fingerprints and StepFinder remain adjacent boundary comparators [7], [8], [9], [10], [11]. Table 2 states the observable eligibility rules.

We use four narrative groupings of objectives and observer roles, not mutually exclusive classes. An unauthorized identity inference can be both attribution and a confidentiality concern. Each record separately specifies the inference target, observer, authorization relation and protected interest: **Type I** concerns confidentiality/privacy; **Type II** concerns model, harness or identity attribution; **Type III** concerns evasion of a platform's human–agent detector; **Type IV** concerns authorized operator diagnostics. Types can overlap. WRA protects research intent (I); UI Traces attributes agents (II); AHB evades a defensive detector (III); ClawGuard diagnoses loops (IV) [6], [5], [12], [13]. These roles are mapped per core family, alongside generating and receiving layers, in Table 6.

TABLE 2. OBSERVABLE-BASED ELIGIBILITY. A HIDDEN TARGET ALONE DOES NOT MAKE A SEMANTIC ATTACK A SIDE CHANNEL.

Observed input	Decision and required distinction
Length, timing, cache response	Core when measured execution dependence supports hidden-target inference.
Own-site actions	Core attribution with site permission; no automatic access to other sites or private tool logs.
Reply states the secret	Direct disclosure; adjacent, not auxiliary-channel evidence.
Reply style/content identifies a user	Adjacent unless a separately evaluated auxiliary mechanism is established. WildAuth stays a mixed-view comparator.
Output-filter membership oracle	Can be core: the filter's execution, not ordinary semantic output inference, mediates the signal.
Privileged loop/fault logs	Separate diagnostic component when C1–C5 hold; does not increase confidentiality evidence.

2.2. Corpus construction and reconciliation

We construct a purposive corpus through 40 seeds, venue-targeted proceedings screening and one-hop citation snowballing, supplemented by database and bulk keyword searches. The supplementary search returned 615 occur-

rences and 565 unique records; auditing 204 exclusions rerouted four records, yielding 15 new core and seven adjacent families. The final collection has 58 core and 24 adjacent families. The Boolean queries recover 25/60 previously indexed families (41.7% seed-set coverage); proceedings and citation routes supply complementary families. The assembled corpus and source versions define the scope of the synthesis. Appendix B provides the complete result-level index.

2.3. Coding procedure and consistency

AI assistance produced initial schema-guided field drafts. Authors checked and corrected all 39 result records and the 180 archived audit cells against source section/table/figure/paragraph locators, and reviewed all 82 family classifications using titles, abstracts and experimental sections. Every record carries a publicly inspectable source locator, enabling readers to check the exact result behind a comparison. The archive retains the initial extraction and author-adjudicated columns, disagreements and subsequent source corrections. Appendix A describes this audit trail.

SoK Takeaway 1 (Map evidence to the receiving deployment). A family-level label does not identify which signal, observer permission or protected objective a result supports. The complete core map separates signal origin from receiving access; the result ledger then binds each comparison to a source version and operating point.

3. Result Records, Comparability Rules and Validity

The atomic unit of our repository is one *evaluated result record*, rather than a whole paper or a single headline score. A result record binds a source version and locator to seven structured fields: Source, Target (and threat polarity Types I–IV), Receiver projection, Intervention boundary, Attacker training/calibration (D_0: clean-condition fit; D_2: protected-condition fit), Workload/holdout, and Outcomes (inference metric, task/component utility, and cost object; Table 3). A study family contributes multiple records whenever it evaluates distinct execution modes (e.g., stored capture vs. live loopback in Dangol [14]), distinct mechanisms (e.g., K-prefix KV simulation vs. GPTCache semantic de-identification in Early Bird [3]), or separately scoped performance benchmarks (e.g., SafeKV [15] and OTRO [16]).

Indexing schema and comparison contracts. The mnemonic SCOPE (Secret/target, Channel, Observer, Pipeline, Exploitation goal) names the target-and-channel columns of our index, paired with evaluation provenance. Generating layers span hardware (P1), serving (P2), transport (P3), application/API (P4), orchestration (P5), tools/MCP (P6), environment (P7) and multi-agent communication (P8). We apply three comparison contracts (Table 4): a comparison is *compatible* when required fields align; *incompatible* when a positive mechanism or boundary conflict is established (such as substituting token-payload ratios for wire bytes or offline timestamp edits for live action delays); and *unresolved* when a required binding is unreported (U), unavailable (A) or unverified (?). Table 5 illustrates a complete record-to-judgment lineage.

TABLE 3. A SINGLE RESULT RECORD. ONLY FIELDS RELEVANT TO THE REQUESTED COMPARISON MUST ALIGN; UNAVAILABLE REQUIRED FIELDS REMAIN UNRESOLVED.

Record field	Required meaning
Source	Family/result ID, version, source locator, reading scope and hash.
Target	Hidden property, inference goal and protection relation: confidentiality, attribution, human/agent distinction or authorized diagnosis.
Receiver	Actual permission and released representation, including session/document linkage and any auxiliary inputs.
Intervention	Where protection ends; whether edits affect stored traces, transmission or executed actions.
Attacker	Fit/calibration condition, test population, training access and adaptation budget.
Workload	Task, system, topology/hardware and grouping/holdout.
Outcomes	Inference metric / denominator; objective or proxy utility; applicable cost object; uncertainty and field states.

TABLE 4. COMPARISON RULES. A KNOWN MISMATCH AND A MISSING BINDING HAVE DIFFERENT CONSEQUENCES.

Requirement	Condition and resulting license
Descriptive comparison	Align target, receiver permission, metric meaning and cost object. Different systems may be described together; token payload and wire bytes cannot be ranked as the same cost.
Paired effect	Compare the same defended test units and outcome. A known test-unit mismatch makes that subtraction incompatible; an unverified fit binding leaves the effect unresolved.
Deployment transfer	Establish a bridge for the changed receiver, intervention, workload, topology or hardware. A demonstrated violation blocks the proposed transfer; an untested bridge leaves it unresolved. Permit only the range and endpoints actually covered.

Executable comparison rules. The record schema and comparison contracts precede the audit results. Table 8 gives all nine operational rules and a shared evidence-status alphabet; Table 9 applies them to every mitigation family. Twenty-four constructed cases and seven mechanism checks exercise rule application in the artifact, while the source-linked cases in §5 explain the substantive comparison judgments.

4. Corpus Map Across 82 Study Families

4.1. Origins, receiving projections and threat polarities

Across all 58 core families in our unified repository, 38 inherit runtime, serving or hardware channels (Origin B), while 20 concern agent-workflow channels arising from

TABLE 5. FILLED RECORD AND ITS LICENSED JUDGMENT. VALUES ARE SOURCE-REPORTED; NO CROSS-MODE EFFECT IS COMPUTED.

Join key / field	Entry
Family / result	Dangol workflow / `DANGOL-livebinding`.
Version / locator	arXiv:2606.07150v3, §9.7, live-protocol paragraph after Table 5 [14].
Target / view	Workflow classification from live Oblivious HTTP (OHTTP)/Agent-to-Agent (A2A) traffic; not GUI-state observation.
Intervention	Live fixed-rate cover with fresh per-session identifiers, distinct from transformed stored captures.
Workload / testbed	Workflow classes with client, relay, gateway and agent co-located on one loopback host.
Outcome / cost	Accuracy 0.30; about 13× bytes and 11× latency in that condition.
Unresolved fields	Objective terminal task utility U; precise paired live-refit comparison ?.
Licensed judgment	These costs describe this live protocol condition. They neither price dummy GUI actions nor prove task preservation. A matched objective endpoint is needed for the latter.

orchestration, tool dispatch, browser navigation and multi-agent protocols (Origin A; Table 6). Signal-generation origin is distinct from receiver location and threat polarity. The full map makes these distinctions visible family by family.

Cross-origin synthesis. Origin B includes token/packet lengths, cache timing and co-resident hardware channels; Origin A includes visited domains, tool dispatch and browser actions. Receiver permissions cut across these origins: a remote network observer, a same-site script and a privileged diagnostic hook expose different views even when they share an attribution target. Thus Origin A/B cannot substitute for Types I–IV or observer permissions. WRA's read-only privacy decoys, AHB's stateful evasion and SplitDecoy's exposed return path illustrate transfers that fail when these axes are collapsed [6], [12], [32]. The map retains every core family rather than using these examples as a substitute for coverage.

Layer incidence highlights where the systematization's comparison problem arises. In the 58-family map, signal-generation labels cluster at P2 serving/runtime (32 families) and P5 orchestration (19); receiving views cluster at P4 application/API (18), P1 hardware (16), and P3 transport (14). Generation at P3/P4 is rare (one family each), despite frequent observation at those layers. This source/receiver asymmetry makes the transfer question concrete: protection at a frequent source layer cannot be presumed to cover a different receiving projection. The map records layer sets, so it does not assign a causal edge to each source/receiver pair.

Purpose labels further show why a single "agent side channel" category is insufficient: Type I appears 41 times, Type II 13, Type III 2, and Type IV 5. The 32 map-only families extend coverage beyond mitigation comparisons: 15 list serving origins, 11 hardware and 11 orchestration; 19 concern confidentiality, nine attribution, one evasion and all five authorized-diagnostic families. They comprise 22 families without a scoped mitigation evaluation, five diagnostics, four unbound quantitative assertions, and SplitDecoy's boundary case (§6). Mapping these families locates privacy targets, operator capabilities and candidate validation environments even when no scoped mitigation record enters the ledger. Counts are overlapping family-label incidences; the audit then applies its reuse rules to the 39 scoped result records.

4.2. Relationship to prior systematizations

Prior surveys map compound-system threats, TEE boundaries and privacy contracts [64], [65], [66]. WF already supplies feature leakage analysis, cost trade-offs and environment-shift tests [67], [68], [69], [70]. Mathews et al. re-evaluate nine defenses with stronger attacks and an on-network DynaFlow implementation [71]. Adaptive attacks and online-cost evaluation are established foundations.

What this synthesis adds. Our unit is a *scoped result*, so execution mode, protected state and cost remain attached when one paper reports several mechanisms. The 39-record census exposes these within-paper substitutions; the 82-family holdout matrix separates evidence for each transfer axis. The empirical additions are target-dependent calibration transfer, the cost/residual-inference boundary of write-local shaping, and fresh planner behavior after executed responses are withheld by deadlines. WF's packet-shaping costs alone do not measure these tool-state and replanning outcomes. Exps 1–2 instantiate known mechanisms; Exps 3–4 measure their agent-specific operating points and observation boundaries. This is a measured extension of established evaluation practice, not a claim to have discovered adaptive attacks or distribution shift.

5. Three Synthesis Lenses over the Result Ledger

We select examples retrospectively by the *comparison decision*, not the size of a reported effect: a changed receiving path, cover with different state effects, or an adaptation/cost mismatch within a named mechanism. NetEcho/SplitDecoy, WRA/AHB and the component-cost records instantiate these three lenses. They are explanatory examples, not a representative sample. The census in §6 covers all 39 records; the RQ3 matrix covers all 82 families, including those outside the mitigation ledger.

5.1. Case 1: bind the final target-bearing emitter

The downstream re-emission boundary. NetEcho's upstream packet-size shaping is an Origin B result; it does not bind the Origin A tool-call sequence. Its plaintext-aware frontend then emits a new Origin B trace (grouped token counts or character lengths), reopening a size channel after upstream protection [36]. SplitDecoy is a separate return-path record: the cloud sees nonzero gradient row support on every tested frame across nine seeds [32, Sections 5.1–5.3]. The inference is a changed receiver boundary, not one transferable score across B and A.

TABLE 6. COMPLETE MAP OF THE 58 CORE STUDY FAMILIES. EACH FAMILY APPEARS ONCE; LAYER AND TYPE LABELS MAY BE MULTIPLE AND DO NOT IMPLY ONE-TO-ONE CAUSAL PATHS. THE BLOCKS GROUP SOURCE LAYERS, NOT EVIDENTIAL QUALITY.

Core family	$G \to O$	Type
Origin B: hardware generation, P1 (8)		
Edge EM leakage [17]	$1 \to 1$	I
Energon [18]	$1 \to 1$	II
MoEcho [19]	$1 \to 1$	I
I Know What You Said [20]	$1 \to 1$	I
Behind Bars [21]	$1 \to 1$	I
PAGE404 [22]	$1 \to 1$	I
Spill the Beans [23]	$1 \to 1$	I
ArchSnoop [24]	$1 \to 1$	II
Origin B: hardware and service, P1/P2 (8)		
SparLeak [25]	$1,2 \to 1$	I
Shape and Substance [26]	$1,2 \to 1$	I
TDXRay [27]	$1,2 \to 1$	I
SparSEEty [28]	$1,2 \to 1$	I
OTRO [16]	$1,2 \to 1$	I
Detokenization Leaks [29]	$1,2 \to 1$	I
LLM-WFIN [30]	$1,2 \to 1,3$	I
LLM-serving RCA [31]	$1,2 \to 2$	IV
Origin B: service generation, P2 (13 of 22)		
SplitDecoy [32]	$2 \to 2$	I
KVGov [33]	$2 \to 2,4$	I
LLMs Have Rhythm [34]	$2 \to 3$	II
Remote timing [35]	$2 \to 3$	I
NetEcho [36]	$2 \to 3$	I
Prompt Echo [37]	$2 \to 3$	I
Speculative decoding [38]	$2 \to 3$	I
Remote keylogging [1]	$2 \to 3$	I
Whisper Leak [39]	$2 \to 3$	I
Cache contention [40]	$2 \to 4$	I
Routing metadata [41]	$2 \to 4$	I
SafeKV [15]	$2 \to 4$	I
ML privacy channels [42]	$2 \to 4$	I

Core family	$G \to O$	Type
Origin B continued: service generation, P2 (9)		
Early Bird [3]	$2 \to 4$	I
Prompt-cache audit [43]	$2 \to 4$	I
InputSnatch [44]	$2 \to 4$	I
Leaky Language Models [45]	$2 \to 4$	I
PromptPeek [2]	$2 \to 4$	I
SpliceLeak [46]	$2 \to 4$	I
Time Will Tell [47]	$2 \to 4$	I
CachePrune [48]	$2 \to 4$	I
MoE prompt stealing [49]	$2 \to 4$	I
Origin A: cross-layer sources (5)		
ClawGuard [13]	$1,5 \to 1$	IV
MIRAGE [50]	$2,5 \to 2$	IV
Cloud traffic benchmark [51]	$2,5,8 \to 3$	I,II
Web-agent unmasking [52]	$3,7 \to 3,7$	II,III
LumiMAS [53]	$4,5 \to 4,6$	IV
Origin A: orchestration/workflow sources (15)		
AgentPrint [4]	$5 \to 3$	I,II
Cycle detection [54]	$5 \to 4,6$	IV
SIGLEAK [55]	$5,6 \to 4,6$	I
ProcGrep [56]	$5,6 \to 4,6$	II
AgentProv [57]	$5,6 \to 6$	II
Lidar [58]	$5,6 \to 6$	II
Dangol workflow [14]	$5,6,8 \to 3,6,8$	I
Campaign attribution [59]	$5,6,8 \to 6,8$	II
WRA [6]	$5,7 \to 3$	I
PersonaFingerprint [60]	$5,7 \to 3$	I
Whose Agent? [61]	$5,7 \to 7$	II
AgentTell [62]	$5,7 \to 7$	I
UI Traces [5]	$5,7 \to 7$	II
FP-Agent [63]	$5,7 \to 7$	II
AHB [12]	$5,7 \to 7$	III

G denotes signal-generation layers and O the receiving observation layers. Layer indices: 1 hardware; 2 serving/runtime; 3 transport; 4 application/API; 5 orchestration; 6 tools; 7 browser/environment; 8 multi-agent communication. Types: I confidentiality/privacy; II attribution; III human–agent distinction/evasion; IV authorized diagnostics. These are objectives and observer roles, not mutually exclusive security guarantees. Privileged operator receivers and physical receivers are marked by the layer they inspect. Source versions, locators and normalization of prose-only layer descriptions are in the artifact.

The data processing inequality requires $S \to Z \to Y$: target-oblivious postprocessing cannot increase $I(S;Z)$ [72, Theorem 2.8.1]. Plaintext-aware re-emission or backward computation changes that premise. The synthesis locates this premise change in published records; Exp 1 measures its streaming cost consequences.

5.2. Case 2: resolve the WRA versus AHB cover asymmetry

Two axes of asymmetry: threat polarity and action risk. WRA's one-decoy E_{func} drops from 0.770 to 0.610, whereas AHB's Trip Planning success falls from 0.75 to 0.15 under `Online+Fake` humanization [6], [12]. Grouping both as "cover traffic" conceals different objectives and executions.

WRA discards decoy reads for Type I confidentiality; AHB executes GUI events for Type III evasion, potentially navigating or submitting forms. Their utility outcomes are not contradictory; AHB's missing condition-specific denominators also preclude a matched causal effect. Shared Origin A does not establish shared state effects.

Terminal-state reward and dispatch-time authorization answer different questions: an unconfirmed write can still reach the intended final state. Exp 2 (§8.2) checks both; the fresh-planner study measures the separate deadline/utility boundary.

5.3. Case 3: bind adaptive retraining and cost objects

The $D_0 \to D_2$ rebound and component-cost boundaries. A matched adaptation contrast compares clean-trained D_0 and defense-trained D_2 on identical defended tests. This follows established adaptive WF evaluation; the rebound itself is not our discovery. UI Traces and Dangol identify the additional record-binding question: whether a refit, measured delay and utility refer to the same executed condition

[5], [14]. Exp 4 tests that question with timing and matched observation sets.

Early Bird's 6ms/4% GPTCache cost does not price K-prefix isolation; KVGov's 1.3 μs HMAC microcost omits lost reuse; SpliceDefense's QCP cost does not price CTBF+QCP [3], [33], [46]. CachePrune binds protection, QA/summarization utility and throughput, while OTRO separates conditional leakage from TTFT/padding costs [48], [16]. The 39-record census makes these component substitutions observable without ranking incompatible outcomes.

6. Comparability Audit Across All 58 Core Families

6.1. Unified 58-core disposition and 39-record ledger

The corpus partitions as $82 = 58$ core $+24$ adjacent families. Within the core, $58 = 20$ mitigation $+6$ contextual-record $+32$ map-only families; the result ledger contains $39 = 33 + 6$ records, respectively, from the first two groups. The six are SIGLEAK, AgentTell, FP-Agent, Behind Bars, LLMs Have Rhythm and ML privacy channels [55], [62], [63], [21], [34], [42]. They supply fixed-guardrail, access-control, interference, VPN or filter-origin contrasts rather than the audited mitigation claim. The remaining 32 comprise 22 without a scoped mitigation evaluation, five diagnostics, four unbound assertions and SplitDecoy's boundary case. Appendix B lists all 39 records with source locators; Table 6 covers all 58 families. Table 7 groups mechanisms; Table 9 reports the 20-family audit.

Whole-ledger synthesis. The complete census assigns 20 records to executed collection/prototypes, six to trace transformation, five to simulation/estimates, six to component/calibration measurements and two to unresolved placement (per-record assignments in the supplement). These mutually exclusive modes explain why a mechanism name cannot carry utility or cost between records. Executed collection establishes its stated execution scope, not automatically full-workflow utility. Table 7's eight rows group mechanisms; they are not eight sampled studies.

6.2. Within-family and within-cohort record comparisons

While the machine-readable audit retains all $\binom{33}{2} = 528$ baseline pairs for completeness, 426 pairs cross unrelated mechanism cohorts and are outside scope by construction. The 13 within-family pairs are included in the 102 within-cohort pairs: $528 = 426 + 102$, with no additional subtraction of 13. Focusing on the **13 within-family pairs** (multiple records inside the same paper) and **102 within-cohort related pairs** (66 traffic-envelope, 28 cache-service, 3 behavioral-humanization, 3 generator-fingerprint, 1 local-memory, 1 fixed-instruction) reveals why paper-level comparisons fail:

- Even within the **13 same-family pairs** (plus 2 same-family pairs in the additions: `CACHEPRUNE-selective` vs. `CACHEPRUNE-context` and `OTRO-unpadded` vs. `OTRO-epochpad`), nearly half (6/13 baseline; 7/15 full) measure different primitives, targets or cost objects (e.g., Early Bird's K-prefix oracle simulation vs. its GPTCache semantic prototype; SpliceDefense's CTBF signal suppression vs. QCP-only prefill cost; OTRO's unpadded TTFT vs. derived epoch-padding cost).
- Among the seven aligned-outcome baseline same-family pairs, four evaluate different collection or execution populations (e.g., Dangol's stored-capture feature ablation at 0.17 accuracy vs. live loopback OHTTP at 0.30 accuracy) and three lack a verified matched-test binding, so none licenses cross-record subtraction as a single paired deployment effect.
- Across the **102 related-cohort pairs**, only 16 (15.7%) support a source-qualified descriptive comparison, whereas 80 (78.4%) exhibit positively established receiver, target or cost-object mismatches and 6 (5.9%) remain unresolved.

6.3. Nine-item split reporting audit across 20 mitigation families

Table 8 defines each item before its application to the 180 cells in Table 9. Applicability follows the result's reuse claim: a fixed analytical attack can make adaptation inapplicable, while a component benchmark can support component utility. U and ? retain distinct source states. The 11 unresolved adaptation cells comprise five R4 and six R5 bindings, explicitly distinguished from an attack failure.

For Dangol, R5 is Bound for in-fold stored-capture experiments; the live-refit paired comparison in Table 5 remains ?. These are different records, so we retain the family count of two and explicitly restrict its meaning.

SoK Takeaway 2 (Evidence needed for reuse). The audit distinguishes component-or-task utility (Bound 3/17 applicable families) from deployment-task utility (2/16 when TDXRay's lookup microbenchmark is treated as inapplicable); these are scope sensitivities, not quality ranks. R5 binds matched adaptation for 2/11 applicable families; a narrower learned-adaptive-attack denominator excluding Shape and Substance is 2/10. CachePrune binds inference, utility and cost together; it does not thereby establish adaptation. The record index in Appendix B identifies the underlying operating points.

7. Systematic Holdout Evidence for Transfer

7.1. Six axes across all 82 families

Appendix D gives the complete family-by-axis matrix, including the 24 adjacent families. It codes task/input/user, model, harness/topology, hardware, network/load and time separately from the archived source-bound extraction. H denotes an explicitly reported held-out unit/domain; V a

TABLE 7. EIGHT INTERVENTION FAMILIES ORGANIZED BY THREAT POLARITY (TYPE I: CONFIDENTIALITY PROTECTION; TYPE II: PROVENANCE/IDENTITY ATTRIBUTION; TYPE III: BOT-DETECTION EVASION), RECEIVING VIEW AND SUPPORTED DEPLOYMENT JUDGMENT. BECAUSE OBJECTIVES AND RISK MAGNITUDES DIFFER, THESE ROWS ARE NOT A COMMON EFFICACY RANKING.

Intervention & polarity	Evaluated view, target and source	Established result and applicable boundary
Padding / batching (Type I: confidentiality)	Network packet lengths and groups; topic inference. NetEcho and Whisper Leak [36], [39].	Collected padding traces retain inference signal; packet-group batching is evaluated by replay. A protected upstream length view does not cover downstream re-emission or residual grouping. Live incremental cost remains U.
Constant-rate / transport cover (Type I: confidentiality)	Network workflow fingerprint; Dangol's OHTTP testbed [14].	The co-located loopback setting yields 0.30 recovery at about $13\times$ bytes and $11\times$ latency. Transformed-capture results are separate; neither result supplies objective agent-task utility.
Concurrent query decoys (Type I: confidentiality)	Destination/domain observer; private prompt and trait inference. WRA [6].	Real/decoy research prompts execute in parallel as read-only queries whose decoy answers are discarded locally; interleaved-domain evaluation reduces one-decoy E_{func} from 0.770 to 0.610. Terminal decoy utility/cost are U.
Action humanization (Type III: bot-evasion)	Mobile GUI touch traces; evading human–agent bot detection. AHB [12].	Unlike WRA's read-only privacy decoys, AHB's `Fake` mode dispatches live mobile GUI touches to evade platform bot detectors, dropping Trip Planning success from 0.75 to 0.15. Non-Fake `History+Long` (0.7143 success) avoids GUI mutation.
Cache isolation / keying (Type I: confidentiality)	Co-tenant prefix timing; SafeKV, KVGov and CachePrune [15], [33], [48].	SafeKV reports post-trigger AUC near 0.5 and separate TTFT/throughput comparisons. KVGov's HMAC median 1.3 μs prices key computation, excluding lost reuse. CachePrune jointly binds marked-token protection, QA/summarization utility and serving performance.
Cache timing equalization (Type I: confidentiality)	Co-tenant cache-fusion timing; SpliceDefense [46].	Constant-Time Boundary Fusion (CTBF) suppresses the tested signal over 15 tasks/75 steps. Quantized Chunk Padding (QCP) adds mean 7.5 tokens and < 5ms/$< 1\%$ prefill time; this does not price combined deployment.
Timestamp perturbation (Type II: attribution)	Recorded website action metadata; model/agent attribution. UI Traces [5].	Fitting on delayed traces (D_2) recovers substantial attribution performance under offline timestamp edits that retain action types/order. Live execution delay and task outcomes are not measured by the trace transformation.
Oblivious / unshared memory (Type I: confidentiality)	Host or local resource observations; TDXRay, Gao et al. and OTRO [27], [20], [16].	TDXRay measures oblivious-map lookup cost; Gao measures loading/memory cost after disabling zero-copy. OTRO reduces conditional access-count information (3.94 to 0.04 bits) with separate unpadded TTFT (4.59%) and epoch-padding (2.8ms) bounds.

Authorized operator diagnostics (Type IV: ClawGuard, cycle detection, LumiMAS, LLM-serving RCA and MIRAGE) are baseline telemetry rather than privacy attacks or evasion defenses, and are cataloged separately.

condition comparison or recalibration; X a scoped restriction; ? unresolved extraction. H does not certify every grouping detail or successful transfer, and ? is never counted as absence. Conceptual families receive an inapplicable dash.

The matrix contains H entries for 13 families on task/input/user, three on model, two on harness/topology and one on network; none is resolved as H for hardware or time. The latter axes contain seven V entries each: calibration, condition sweeps and temporal observations already exist in the literature. With 73 hardware and 71 temporal entries unresolved, this is a map of *documented evidence*, not a claim that those papers omitted such tests. Multiple H entries are counted once per family/axis, not pooled as independent experiments.

7.2. What each holdout licenses

Whisper Leak holds out target prompts; FP-Agent has a separate task holdout but unresolved visitor/host separation. Dangol's leave-composition-out and the traffic benchmark's topology shift change a different generator [39], [63], [14], [51]. LLM-Fin and ArchSnoop test unseen variants/configurations within known model families; LeakyLMs conditions architecture inference on known GPU type and calibration [73], [24], [45]. Thus a positive model holdout does not establish unknown-hardware transfer. UI Traces' recorded question overlap further shows why task labels and actual held-out units must remain distinct [5].

7.3. Operational consequence

Select the axis changed by deployment, require a source-bound separation on that axis, then check whether fitting or calibration crossed the split. Otherwise scope the result to its observed condition. Feature ablations and additional models alone supply V evidence, not a bridge to an unseen generator. This rule also governs our endpoint experiments: within-model refits, source-trained attacker transfer and source-calibration transfer occupy separate matrix cells. The systematic result is the mismatch between the requested transfer axis and the available evidence, not a ranking by the number of holdouts.

8. Testing Defense Transfer Boundaries

Four experiments test the ledger's transfer obligations. Exps 1–2 are mechanism checks of emitter placement and isolated state; their novelty does not reside in DPI, equality by construction or clean-to-defended retraining. Their measurements expose streaming cost and joint state/link constraints. Exp 3 adds target-dependent calibration transfer and economical but incomplete write shaping; Exp 4 tests

TABLE 8. OPERATIONAL RULES FOR THE NINE REUSE BINDINGS. APPLY EACH ITEM TO A SPECIFIED RESULT RECORD AND DEPLOYMENT CLAIM.

Item	Evidence required for a Bound (B) judgment
R1 Target	Hidden property, inference goal and authorized/protected interest are explicit.
R2 Receiver	Observer permission, released representation and auxiliary inputs identify the evaluated view.
R3 Intervention	Placement and execution mode bind the transform to stored data, wire emission or executed actions.
R4 Protected fit	Attacker fitting/calibration uses the protected condition; training and protected testing are identified.
R5 Matched adaptation	Clean- and protected-condition fits are evaluated on the same defended test population.
R6 Workload	Tasks, system/generator configuration and holdout specify the supported transfer axis.
R7 Inference	Metric, denominator and evaluation condition bind the attack outcome to the scoped mechanism.
R8 Utility	Task/component performance is measured at the same protected operating point, with its state boundary explicit.
R9 Cost	Resource units, baseline and measurement object price that operating point.

B: bound; P: partial binding; U: unreported in inspected scope; A: source unavailable; ?: binding unverified; –: inapplicable. First determine applicability, then assign evidence status. Family B means at least one scoped record satisfies the item, not that every result does.

TABLE 9. PER-FAMILY EVIDENCE BINDINGS FOR ALL 20 EVALUATED-MITIGATION FAMILIES. R1–R9 ARE SCOPED REUSE ITEMS, NOT OVERALL PAPER SCORES.

Family	R1	R2	R3	R4	R5	R6	R7	R8	R9
UI Traces [5]	**B**	**P**	**B**	**B**	**B**	**B**	**B**	U	–
WRA [6]	**B**	**P**	**B**	–	–	**P**	**B**	**P**	U
NetEcho [36]	**B**	**B**	**B**	**B**	**P**	**B**	**B**	U	U
Early Bird [3]	**B**	**B**	**B**	**B**	**P**	**B**	**B**	U	**B**
Whisper Leak [39]	**B**	**B**	**P**	?	?	**B**	**B**	U	**B**
SpliceLeak [46]	**B**	**B**	**B**	–	–	**B**	**B**	U	**B**
Time Will Tell [47]	**B**	**B**	**B**	?	?	**B**	**B**	U	**B**
I Know What You Said [20]	**B**	**B**	**B**	–	–	**B**	U	–	**B**
Speculative decoding [38]	**B**	**B**	**P**	?	?	**B**	**B**	–	**B**
Dangol workflow [14]	**B**	**B**	**B**	**B**	**B**	**B**	**B**	U	**B**
Cloud traffic benchmark [51]	**B**	**P**	**B**	?	?	**P**	**B**	**P**	U
Remote timing [35]	**B**	**B**	**B**	–	–	**B**	**B**	U	**P**
TDXRay [27]	**B**	**B**	**B**	–	–	**B**	U	**B**	**B**
SafeKV [15]	**B**	**B**	**B**	–	–	**B**	**B**	U	**B**
AHB [12]	**B**	**B**	**B**	**P**	?	**B**	**B**	**B**	U
KVGov [33]	**B**	**B**	**B**	–	–	**B**	**B**	U	**B**
CachePrune [48]	**B**	**B**	**B**	–	–	**B**	**B**	**B**	**B**
OTRO [16]	**B**	**B**	**B**	–	–	**B**	**B**	–	**P**
SparLeak [25]	**B**	**B**	**B**	?	?	**B**	**B**	**P**	U
Shape and Substance [26]	**B**	**B**	**P**	U	U	**B**	**P**	U	**P**
Bound	20	17	17	4	2	18	17	3	11
Applicable	20	20	20	11	11	20	20	17	19

B: bound; **P**: partial; U: unreported in inspected scope; ?: unverified binding (11 cells); –: inapplicable. Table 8 defines R1–R9 and the shared status alphabet. A bound cell is supported by at least one scoped record. Dangol R5 concerns stored capture; its live-refit binding remains unverified. CachePrune jointly binds R7–R9 at one operating point. Source follow-up resolves cloud-traffic R3 to B and Shape and Substance R4/R5 to U.

the attack-suite and measurement-mode boundary. Table 10 separates populations; Appendix C indexes all studies.

Shared measurement conventions. A byte-transparent relay captures post-handshake TLS application records on loopback; TLS 1.3 session tickets are disabled. Standard-prompt results use the arm called `baseline` in the artifact; *undefended* denotes transport without a program defense. Balanced accuracy (BA) averages the two class recalls. We report the maximum over each specified attack suite and recompute that maximum in each of 2,000 common bootstrap draws, grouping tasks for reference-program tests and customers for real-agent tests. The original protocol additionally screens for an upper 95% BA bound below 0.60 and maximum AUROC below 0.65. We retain this engineering threshold, reporting its interpretation once in §9.3.

TABLE 10. COHORT, OBSERVATION AND COST DENOMINATORS FOR EXPS 3–4. STANDARD-PROMPT PAIRS: CANCEL/RETURN BA (%). COUNTS COVER BOTH PROMPTS; BYTE RATIOS PRICE JOINT CONSTANT SHAPING AGAINST EACH STUDY'S OWN UNDEFENDED SERIALIZATION.

Study	Observation / fit	Raw BA	Bytes×
Exp 3 frozen	Replay lengths/order; frozen suite	97.2/94.0	15.39
Exp 3 cross-fit	Analytic lengths/order; 11-model cross-fit	91.7/85.5	19.76
Exp 4	New RPC envelope; lengths/order/time	91.7/86.4	15.11

Frozen-split and RPC rows: 276 test episodes, 24 customers, 5/7 positive customers. Exp 3 cross-fit: 690 episodes, 53 customers, 14/18 positive customers. Each prompt has half the episodes. Exp 3 cross-fit fold calibration yields larger templates (5.04× calls); frozen-split and RPC rows use 3.83×. Repeats measure execution variation within these customer groups.

8.1. Exp 1: bind protection and cost to the emitter

Buffered tests use brief/expanded saved responses for 24 topics (12 training, 12 test). Padding only the upstream link leaves verbosity 100% separable at a plaintext-aware terminal emitter; padding both reduces BA to 50%, reproducing NetEcho's mechanism [36]. A training-fitted buffered cap achieves 50% at 2.18× wire bytes under TLS 1.2 (2.19× under TLS 1.3).

Streaming the same 48 texts as per-token server-sent events changes the cost contract [1], [39]. Per-event padding retains token count and 100% verbosity inference. Eight-token batching cuts bytes to 0.21× and clean-trained BA to 50%, yet defended-data refitting restores 100%. Constant shape reaches 50% at 2.42× bytes and 208 versus 105 token intervals; two overflowing test outputs remain identifiable (Table 11). **Finding.** The same saved content yields different protection and delay under buffering and streaming. A buffered cap's byte ratio cannot price a streaming deployment, whose first-output delay and emission schedule are different outcomes.

TABLE 11. EXP 1: THE SAME 48 SAVED RESPONSES EMITTED BY A BUFFERED OR A TOKEN-STREAMING FRONTEND (TLS 1.3 LOOPBACK CAPTURES). BYTES ARE RELATIVE TO THE UNPADDED EMITTER OF THE SAME KIND. TTFB/DONE: FIRST AND LAST DELIVERED BYTE IN TOKEN INTERVALS (ACCOUNTING PROXY). ACCURACIES ON 24 HELD-OUT OUTPUTS (12 TOPICS); ID: CLOSED-WORLD TOP-1 IDENTIFICATION AMONG 48 RESPONSES (CHANCE 2.1%).

				Verbosity		Prefix	
Emitter / framing	Bytes×	TTFB	Done	D_0	D_2	D_2	ID
Buffered, unpadded	1.00	105	105	100.0	100.0	91.7	87.5
Buffered, training cap	2.19	105	105	50.0	50.0	50.0	2.1
Stream, per token	1.00	1	105	100.0	100.0	87.5	100.0
+ random event pad	1.29	1	105	100.0	100.0	87.9	62.3
+ constant event size	1.23	1	105	100.0	100.0	87.5	58.3
8-token batches, const.	0.21	8	105	50.0	100.0	87.5	20.8
Const. events + count cap	2.42	1	208	50.0	50.0	50.0	6.3

D_2 is the best of total-byte threshold, record-count threshold and a logistic sequence model, each fitted on defended training outputs. Random padding: 10 socket-measured draws plus an independent profiling run. TLS 1.2 gives identical accuracies and byte ratios within 0.03. Verbosity contrasts brief/expanded outputs; Prefix contrasts the first two/three sentences of each expanded output. Tokens use o200k_base. Training count caps are 208/88 events, with 2/24 and 5/24 test overflows, respectively.

8.2. Exp 2: compose state preservation with link protection

Execution boundary. Among 115 public retail tasks, 97 have an initially pending order for the task user. Three order-selection seeds yield 291 cover placements before the reference program. Same-item modification changes status and appends a refund in 173 placements despite unchanged item and price; the other 118 calls fail the variant-availability check. Idempotence, $f(f(s,a),a) = f(s,a)$, therefore differs from state preservation, $f(s,a) = s$. Cancelling another user's order changes the shared database. The shadow executor isolates cover effects on a separate state and suppresses its responses; a live-execution control exposes the resulting side effects.

The dispatch observer sees ordered tool names and response lengths. Policies add cancellation, pad its count, interleave donor programs or equalize all tool counts; donors/templates use the known 115-program population. Task-grouped five-fold attacks fit clean/defended gradient boosting and logistic regression on counts, bigrams and lengths, with a count-rule baseline and three seeds (18/32 positive cancellation/return tasks). One inserted cancellation drives clean-trained BA to chance, while refitting restores 100%. Count equalization removes that rule's signal, but correlated tools retain 76.9% BA. Donor programs, inspired by WRA's discarded retrieval cover [6], leave 83.5–91.9%. Shape equalization reaches 50.0/54.5% at 6.7 extra calls per real call (Table 12). Shadow checks match the original live state in every case; live cancellation changes it in every case.

Final network view. Four cover policies and three framing choices produce 4,140 TLS 1.2 sessions. The network observer receives direction-tagged record lengths and order. A training-fold profiler maps request sizes to tool types before applying dispatch attacks: request-only identification reaches 89.9/91.4% (582 calls; majority 29.4%). Per-call buckets leave return BA at 73.9%; joint shape cover and constant framing reduce cancellation/return to 54.4/50.0%, at 26.3× bytes and 7.7× calls.

TABLE 12. EXP 2 MAIN DISPATCH RESULTS: ALL 115 REFERENCE PROGRAMS, THREE SEEDS, KNOWN-POPULATION DEFENSE CALIBRATION; TASK-FOLDED ATTACKERS. MAXIMUM BA (CANCEL/RETURN); EXTRA CALLS PER REAL CALL; DATABASE EQUALITY.

Policy	BA (%)	Calls+	DB eq.
No cover	100.0/100.0	0.00	100%
Cancel, $p = .5$	86.1/100.0	.10	100%
Cancel, $p = 1$, shadow	100.0/100.0	.20	100%
Cancel, $p = 1$, live	100.0/100.0	.20	0%
Cancel count $\to 2$	76.9/100.0	.35	100%
One decoy program	91.9/85.9	1.01	100%
Two decoy programs	83.5/78.1	1.98	100%
Shape equalization	50.0/54.5	6.71	100%

Finding. Call-level cover and per-call framing address complementary observables. Full constant-rate transport can instead hide packet counts from a passive link observer without executing dummy tools [74]; Exp 4 measures that alternative. At a dispatch view after decapsulation, cover requires hidden markers and routing to the trusted isolated executor (§9.1). The intervention must match the receiver and the protected state.

8.3. Exp 3: real-agent signatures and calibrated shaping

Task and collection. Cancellation and return label the reference program's requested operations, `cancel_pending_order` and `return_delivered_order_items`. This operational target tests whether encrypted metadata reveals which service action a task requests. GPT-6.1 Sol/low plans from public retail policies, schemas and actual sandbox observations; a separate context simulates the user. Two prompts and three repeats on 115 tasks yield 690 runs from 53 customers. The rules prompt adds a customer-details read before order access and a read before each write.

Frozen calibration and behavioral control. All 414 training runs (69 tasks) precede freezing shared cover templates, donors and framing caps. The 276 test runs cover 46 tasks from 24 disjoint customers. Prefix-only scheduling inserts training-derived cover, flushing unmet counts at episode end. Four program conditions per prompt give 2,760 local TLS replays with lengths/order attacks; every failure and handoff remains included. Real-call pass-through plus isolated cover preserves the logical response/state projection (Appendix A); reusing trajectories pairs the network interventions on identical model decisions.

Rules raise read-order adherence from 86.2% to 97.1%, while undefended category inference remains above 93%. Native rewards are 110/138 and 105/138 (paired difference

-3.6 points, 95% interval $[-9.5, 1.4]$). Thus compliance and metadata privacy separate in the same task population. Joint cover and framing reach 50.0/50.0% point BA, compared with 97.2/94.0% undefended; their upper bounds are 60.6/65.5%. The paired decreases are 47.2/44.0 points (Table 13). The rules-arm joint return point is 60.6%, identifying prompt/cohort sensitivity.

Customer-grouped exploratory comparison. Five-fold cross-fitting refits templates, donors, caps and all attackers on other customers, pooling the full known cohort (14/18 positive customers). The analytic TLS model matches all 2,760 saved sessions and 120 additional captures. An 11-model union includes a 1D-CNN and k-FP-style forest-leaf learner. Joint constant shaping reduces BA by 31.2/30.0 points (paired 95% intervals [15.3,41.0]/[14.8,39.2]); buckets reduce it by 34.6/32.8 points, at lower byte cost (Table 14).

Locate the observable operation signature. An oracle selects write exchanges using true tool types, then exposes only ciphertext lengths/order. Writes comprise 19.6% of real calls yet retain 91.7/84.9% BA; removing them leaves 62.8/59.9% (Table 15). The target names an operation, so the result locates its metadata signature. It directs protection toward specific exchanges, while residual read inference shows why hiding one named call is insufficient.

Model-specific repeated evaluation. Luna's 150-run extension contributes a same-provider replication; two additional endpoints each contribute 115 standard-prompt runs with 69 training tasks, a defense frozen before test collection, and 46 test tasks (Table 16). Joint shaping lowers return BA by 16.0 points for MiniMax-M3 and 19.4 for GLM; cancellation changes by $+2.1$ and -50.0 points, respectively. Thus return-operation inference recurs under separate within-model fits, while protection depends on model and target. These are within-model contrasts on known customers, using real agent trajectories and local TLS replay. MiniMax-M3 is the served alias (the deployment owner reports a DeepSeek backend); GLM uses a 16,384-token budget versus 4,096 for MiniMax. Neither endpoint exposes an immutable revision. Complete intervals, failures and write localization accompany the replication supplement.

Measure the model-transfer boundary. A post hoc transfer study adds an exchange-count rule to the inherited six attacks. On GLM, moving MiniMax calibration across models raises cancellation BA from 50.0% to 70.8%, while return changes from 67.3% to 50.0% (Table 17). Transfer therefore trades protection between targets; no scalar model ranking captures this boundary. Source-trained attackers also transfer: undefended return BA is 77.5% for MiniMax-to-GLM and 88.1% for GLM-to-MiniMax. These two-endpoint results use known customer partitions, not independent providers or fixed weights. Write-only padding costs 1.45/1.51$\times$ bytes for MiniMax/GLM but leaves return BA at 61.3/61.6% under the expanded suite. A learned length-based write detector has precision 34.9/98.3% and recall 96.3/89.4%, distinguishing oracle localization from implementable inference. Localization thus proposes economical interventions; residual signals and detector errors determine whether their protection transfers.

Test live planner utility at its own boundary. A paired 50-task GLM study runs 100 fresh planner episodes with real isolated prefix-cover execution and emulated response deadlines. Baseline and prefix-shadow succeed on 37/50 and 33/50 tasks (difference -8.0 points, customer interval [-23.9,6.5]). Withheld executed-tool responses trigger subsequent planner actions. This measures the prefix scheduler's utility under injected delays; terminal flush and TLS framing belong to a different mechanism. The supplement retains every request, fault and outcome.

Finding. Better read-order compliance leaves operation inference intact; calibrated joint shaping reduces that fitted-suite inference; and the localization identifies which exchanges carry it. Semantic-attribute generalization is a separate estimand, tested by the small extension summarized in Appendix C.

8.4. Exp 4: timing, latency and matched transport controls

All 690 saved action trajectories execute through a live local tool server under six conditions, producing 4,140 new TLS sessions. Each real response gates the next action; real and shadow calls have separate states. The original 414/276 customer-disjoint split is retained, with attacks fitted and frozen before test capture. Lengths, directions and arrival times measure the tool RPC chain.

A complete tunnel sends 4,096-byte cells in both directions at 2 or 5 ms target intervals; dummy cells execute no tools. A training-sized 133-slot horizon continues after the last response. Serial and two-in-flight shadow schedules use the frozen prefix-only count policy with constant per-call framing. The attack union includes size/order, timing, combined features and learned flush removal. Seven shared learners, including a 1D-CNN and forest-leaf neighbors, also permit a matched comparison with offline website-fingerprinting (WF) adaptations.

Timing reveals distinguishable cover. For standard-prompt serial shadow, adding timing/flush attacks increases cancellation BA from 50.0% to 61.4% (paired increase 11.4 points, 95% interval [3.7,19.6]); return changes from 50.7% to 53.4%. Flush-phase detection reaches 90.4/92.3% for serial/parallel schedules over 15,278 records from 276 test episodes. A separate cover-versus-real request classifier reaches 82.5/96.1% on 7,639 requests.

We then compose the detector with intent inference: a training-fitted detector strips the predicted suffix before classification. Table 19 shows the recovered category signal, with a strongest return point of 60.7%. The full attack union retains both stripped and unstripped traces. This separates the ability to recognize padding phases from the ability to recover a task's operation.

Match cost to the protected boundary. Within measured RPC conditions (M), per-call constant framing uses 3.92$\times$ bytes and 16.6 ms p95, versus serial shadow's 15.11$\times$ and 26.2 ms; its full-union BA is higher (68.6/62.7% versus

TABLE 13. EXP 3 FROZEN: FROZEN-CALIBRATION OPERATION-CATEGORY DIAGNOSTICS ON REAL-AGENT TRAJECTORIES. TESTING COVERS 46 TASKS FROM 24 HELD-OUT CUSTOMERS, TWO PROMPTS AND THREE REPEATS (276 EPISODES). NETWORK BA IS THE MAXIMUM OVER THE FIXED ATTACK SUITE, WITH CUSTOMER-CLUSTER 95% INTERVALS.

Condition	Cancel BA [95% CI]	Return BA [95% CI]	Original reward (%)	Bytes×	Calls×
Undefended	97.2 [90.0, 100.0]	94.0 [85.8, 100.0]	79.7	1.00	1.00
Rules prompt	97.2 [90.0, 100.0]	93.5 [85.0, 99.5]	76.1	1.00	1.00
Padding only	66.1 [50.0, 83.9]	55.5 [50.0, 73.0]	–	4.02	1.00
Shape only	60.8 [55.4, 71.3]	51.9 [50.5, 57.8]	–	4.16	3.83
Joint program	50.0 [50.0, 60.6]	50.0 [50.0, 65.5]	–	15.39	3.83
Rules + joint	50.0 [50.0, 65.1]	60.6 [50.0, 73.3]	–	15.72	3.88

Paired joint-minus-undefended BA differences (percentage points): standard-prompt cancellation -47.2 [-50.0, -33.3], return -44.0 [-48.9, -27.8]. These post hoc contrasts use the same customer draws and reselect the attack-suite maximum in each condition. Joint = online shadow cover plus constant framing; padding/shape ablations use standard-prompt trajectories. Ratios use undefended transport within each prompt. Shadow isolation preserves the logical planner transcript; the four program conditions replay actual LLM trajectories over local TLS 1.2. Reward is reported only for original model executions; dashes denote no new live defended-planner reward measurement. Program transforms check logical conformance; see the utility scope in §9.3.

TABLE 14. EXP 3: EXPLORATORY CROSS-FIT COMPARISON UNDER THE EXPANDED 11-ATTACKER UNION: ALL 690 EPISODES AND 53 CUSTOMERS. MAXIMUM NETWORK BA WITH CUSTOMER-BOOTSTRAP 95% INTERVALS. STANDARD PROMPT UNLESS MARKED RULES; BYTE/CALL RATIOS RELATIVE TO UNDEFENDED TRANSPORT.

Configuration	Cancel BA (%)	Return BA (%)	B×/C×
Undefended	91.7 [82.1, 100.0]	85.5 [76.7, 93.4]	1.0/1.0
Joint constant	60.5 [52.9, 72.4]	55.5 [51.8, 64.2]	19.8/5.0
Joint buckets	57.1 [50.0, 71.5]	52.7 [50.0, 61.7]	12.4/5.0
Rules + buckets	56.9 [50.2, 71.2]	52.7 [50.0, 65.0]	12.4/5.0

Joint constant is Exp 3 frozen's shadow-count plus constant-framing design, refit within each fold. Joint buckets changes framing to 1024-byte buckets. Every bootstrap draw reselects the complete maximum. Intervals condition on fitted predictions.

TABLE 15. EXP 3 ORACLE LOCALIZATION UNDER THE 11-ATTACK UNION. STANDARD PROMPT: 345 CROSS-FITTED EPISODES, 53 CUSTOMERS. MAXIMUM NETWORK BA (%) WITH CUSTOMER-BOOTSTRAP INTERVALS.

Target	Full trace	Writes removed	Writes only
Cancel	91.7 [82.1, 100.0]	62.8 [54.6, 75.3]	91.7 [82.1, 100.0]
Return	85.5 [76.7, 93.4]	59.9 [52.2, 71.8]	84.9 [78.8, 94.0]

TABLE 16. EXP 3 MODEL-SPECIFIC REPEATED EVALUATIONS: NETWORK SUITE-MAX BA (CANCEL/RETURN, %). EACH MODEL HAS 46 TEST TASKS AND 24 CUSTOMERS. ENDPOINT ROWS USE THE EXPANDED SEVEN ATTACKS; LUNA RETAINS ITS ORIGINAL SUITE. LUNA USES 29 TRAINING TASKS; THE TWO ENDPOINT REPLICATIONS USE 69. CONTRASTS ARE WITHIN MODEL; ENDPOINT ALIASES AND INFERENCE BUDGETS DIFFER.

Model	Undefended	Joint
Luna	64.2 / 84.9	60.4 / 57.0
MiniMax-M3	58.3 / 83.5	60.4 / 67.5
GLM-5.3-flash	100.0 / 86.6	50.0 / 67.3

TABLE 17. DEFENSE CALIBRATION TRANSFER ON 46 TARGET-MODEL TEST TASKS. ENTRIES ARE CANCEL/RETURN BA (%) UNDER THE EXPANDED SEVEN-ATTACK SUITE; TARGET-MODEL ATTACKERS ARE REFITTED ON TRAINING CUSTOMERS. SOURCE TEMPLATES AND CAPS USE SOURCE-MODEL TRAINING DATA ONLY.

Calibration source	MiniMax target	GLM target
MiniMax	60.4 / 67.5	70.8 / 50.0
GLM	50.8 / 63.0	50.0 / 67.3

61.4/53.4%). This establishes a measured cost–inference trade-off. Measured real-chain p95 latency is 26.2/20.8 ms for serial/two-in-flight shadow, versus 56.4/112.3 ms for 2/5 ms tunnels. Sessions preserve real responses and final live state; fixed tunnel sizes/counts still expose host-scheduled direction/order. Table 18 keeps measured RPC times separate from offline schedule transformations.

Under the common seven attacks, offline random-donor Walkie-Talkie (WT, S) reaches 50.0/56.7% BA at 2.04× bytes; measured serial shadow (M) reaches 50.0/52.9% at 15.11×. This descriptive, shared-metadata comparison identifies a lower-byte shaping candidate. Establishing its online trade-off requires RPC measurements and the extended timing/flush suite; the offline point is neither a theoretical byte lower bound nor demonstrated deployment superiority.

The WF adapters use the same saved traffic and training customers. WT samples donor traces without symmetric site pairing; WTF-PAD samples training inter-arrival histograms; Tamaraw and FRONT supply fixed-rate and front-loaded controls. Input-derived fragments preserve sizes, directions and order, enabling a controlled metadata comparison. Their simulated spans and measured RPC latency remain separate cost objects.

Robustness under WAN transit delay and jitter. Because local loopback exhibits zero transit jitter, we repeat the evaluation on all 276 test episodes under network emulation matching wide-area links (20 ms delay with 5 ms jitter, and 50 ms delay with 10 ms jitter). In the supplementary WAN evaluation, flush-phase detection remains remarkably robust: the frozen loopback detector D_{loop} transfers directly to WAN traffic with 89.4% BA and 0.962–0.963 AUROC (versus 90.4% and 0.967 on loopback), and locally adapted retraining yields 89.0–89.2% BA. The macroscopic density transi-

TABLE 18. TIMING-AWARE EXP 4 COMPARISON: STANDARD PROMPT, 138 HELD-OUT EPISODES, 24 CUSTOMERS (5 CANCEL-POSITIVE, 7 RETURN-POSITIVE). EACH BA PAIR IS CANCEL/RETURN (%). THE COMMON SEVEN MODELS PERMIT A MATCHED-SUITE COMPARISON; THE FULL UNION ALSO RETAINS ALL EARLIER ATTACKS.

View	Condition	Common-seven BA	Full-union BA	Upper BA	Bytes×	Calls×	RPC p95 (ms)
M	Undefended	84.3/76.9	91.7/86.4	100.0/97.2	1.00	1.00	15.4
M	Per-call constant	53.2/59.1	68.6/62.7	92.8/79.5	3.92	1.00	16.6
M	Serial shadow	50.0/52.9	61.4/53.4	71.4/65.8	15.11	3.83	26.2
M	Two-in-flight shadow	50.0/57.3	51.7/57.7	69.1/69.7	15.11	3.83	20.8
M	Full tunnel, 2 ms	59.9/55.2	62.5/55.2	70.2/62.0	125.71	1.00	56.4
M	Full tunnel, 5 ms	54.6/53.7	54.6/57.3	72.9/64.2	125.71	1.00	112.3
S	Record buckets	53.1/69.8	53.1/69.8	78.5/86.9	2.25	1.00	–
S	Tamaraw 40/12 ms	55.3/55.3	55.3/55.3	68.7/64.4	12.15	1.00	–
S	Tamaraw 4/1.2 ms	57.8/51.8	57.8/51.8	71.8/61.8	11.99	1.00	–
S	FRONT	59.0/69.3	59.0/69.3	69.9/81.8	2.51	1.00	–
S	WTF-PAD adaptation	51.9/62.7	51.9/62.7	64.1/79.6	1.88	1.00	–
S	WT random-donor	50.0/56.7	50.0/56.7	67.0/67.8	2.04	1.00	–
S	FRONT, short calibration	52.4/56.0	52.4/56.0	61.3/71.1	2.33	1.00	–

M: measured local tool RPC on saved actions; S: offline traffic transformation. RPC p95 ends at the last real response, excluding LLM/WAN. S-row simulated spans are not RPC latency. Upper BA is the full-union 95% bound. M unions contain 17–22 attacks; S unions contain the common seven. Ratios use prompt-specific undefended captures. The utility scope is defined in §9.3.

TABLE 19. EXP 4 INTENT BA AFTER LEARNED FLUSH REMOVAL (95% CUSTOMER-CLUSTER INTERVAL). EACH ROW: 138 TEST EPISODES, 24 CUSTOMERS. MAXIMUM OF THREE CLASSIFIERS PLUS MAJORITY; THE COMPLETE ATTACK UNION ADDITIONALLY KEEPS UNSTRIPPED OBSERVATIONS.

Prompt	Schedule	Cancel	Return
Standard	Serial	57.8 [50.0, 70.7]	53.4 [50.0, 64.6]
Standard	Parallel	50.0 [50.0, 64.0]	50.0 [50.0, 56.7]
Rules	Serial	50.1 [50.0, 55.6]	60.7 [50.8, 70.9]
Rules	Parallel	50.8 [50.0, 61.7]	54.0 [50.0, 67.0]

tion of dummy cover requests is largely preserved across transit variance. Concurrently, transit jitter degrades the attacker's fine-grained intent inference: under serial shadow cover, full-suite cancellation BA drops from 61.4% on loopback to 50.0% (chance) under 20 ms/5 ms jitter and 54.0% under 50 ms/10 ms jitter. Crucially, WAN emulation exposes an operational failure boundary for fixed-rate tunnels: while loopback completed 276/276 sessions within the fixed 133-slot (266 ms) horizon, network round-trip delays exhaust the slot budget for multi-call agent workflows before all real tool responses return, failing 69.2% (85/276 completed) and 98.2% (5/276 completed) of sessions. In contrast, shadow cover and constant framing preserve 100% correctness and live state fidelity, while their real-chain p95 latency expands proportionally to the serial sum of network RTTs (855.4 ms and 1999.5 ms).

Finding. A length-only verdict misses schedule leakage. The common outcome is suite-maximum BA, always indexed by its observed features and fitted attacker set: Exp 3's 50.0/50.0% and Exp 4's 61.4/53.4% cross an attacker-observation boundary and are not interchangeable protection estimates. Within M, per-call framing costs fewer bytes and less RPC time than serial shadow but leaves higher full-union inference; S identifies lower-byte candidates under the common seven attacks. For a passive ingress-link observer, lightweight shaping is a relevant candidate; for a restricted dispatch observer, execution isolation addresses an additional state boundary. This is the audit's placement-and-cost distinction demonstrated on one workload.

Synthesis. The demonstrations operationalize the audit: match the emitter before borrowing a byte cost (Exp 1), verify executed state before borrowing a cover strategy (Exp 2), bind targets and calibration to a shared cohort (Exp 3), and price timing protection against matched transport alternatives (Exp 4). They show how source-bound comparisons change a concrete deployment decision.

9. Implications, Ethics and Limits

9.1. Deployment paths and reporting obligations

A passive agent–tool observer can be an access-network operator, an authorized VPC flow monitor, or a relay on an inter-organization path. Each position defines a specific observable projection. Table 20 states the corresponding visibility and trusted endpoint. Encrypted ClientHello hides the inner server name when supported, but destination IPs and downstream connections can still identify tools [75]. Our measurements use a single ingress endpoint.

Shadow state has three possible implementations. **(1) Authorized reads and verified no-ops** use the backend with discarded responses; idempotence alone does not preserve state. **(2) Provider-supported isolation** includes Kubernetes server dry-run, separate Stripe sandboxes and transactional rollback [76], [77], [78]. These have different semantics: for example, sequence changes can survive rollback [79]. **(3) A trusted tool gateway** authenticates cover markers and routes shadow writes to an isolated executor. Our local measurements implement the gateway path; the provider-specific paths identify deployments for follow-up measurement.

The trusted gateway/executor can identify cover. Protection ends at exposed markers, routing branches or commit logs; it does not protect against that principal. On

TABLE 20. DEPLOYMENT BOUNDARY: THE RELEVANT PASSIVE OBSERVER AND WHAT THE MEASURED MECHANISM CAN PROTECT.

Setting	Visible information	Applicability / trusted point
Same-host sidecar	Host trace/process/IPC metadata, if permitted	Link padding alone does not hide host instrumentation; isolate the monitor's access.
Same-VPC gateway	Ingress record sizes, times and addresses	Tunnel or shadow shaping can protect this link; gateway/markers remain trusted.
Remote SaaS/MCP	IP, unprotected SNI, per-endpoint flows and times	Shared ingress needed for tool-identity cover; ECH alone does not hide destination IP.
Cross-organization	Each hop's routing; provider calls/commit log	No guarantee across all hops. A trusted isolated executor is an explicit added assumption.

the ingress link, the inexpensive WF adaptations are relevant competitors. At a restricted dispatch view, hiding call profiles additionally requires marker/routing isolation. The choice therefore follows the required observation boundary. Cover also consumes quotas and shared capacity; admission budgets, backpressure and separate credentials must prevent overload or accidental live writes.

9.2. Ethics and artifact scope

Executions use cloned public benchmark databases and loopback sockets; model requests contain public tasks and simulated interactions. No production account is modified or third-party service probed for a vulnerability. Stakeholders include paying tenants, shared-capacity providers and people affected by state changes. Other-account cancellation is a simulator control. Released inference code can support surveillance; examples use public/synthetic identities and controlled endpoints. Bot-detection evasion has a different authorization boundary [12], [6].

AI assisted candidate-field extraction, implementation, source reconciliation and prose revision. Authors retain responsibility for the source-linked comparison and its interpretation. Authors' source verification is specified in §2.3. Saved-output replay, model-generated trajectories and measured local TLS are distinguished throughout. The artifact archives source locators, decisions, observations, calibration and executable checks. An anonymous submission snapshot must omit identity metadata and development history; the separate submission checklist records the AI-use disclosure and author review required at registration.

9.3. Threats to validity and scope of evidence

Source synthesis. The purposive corpus supports within-corpus comparisons; seed-set coverage is not field-wide recall. Many recent sources are preprints with version-bound claims. AI extraction and author adjudication are dependent stages; agreement measures process consistency. Constructed cases check rule execution. The holdout matrix inherits the archived extraction's scope; unresolved axes require source follow-up. Locators and blind-recoding templates enable that audit.

Statistical scope. The BA/AUROC screen is an engineering threshold, not a security definition. No joint design confirms the two-target absolute criterion; paired reductions establish improvements against specified fitted suites. Intervals reselect the suite maximum but omit retraining uncertainty. The frozen split has 5/7 positive customers; the full cohort has 14/18. Repeats add execution coverage. Reference calibration uses the known population. Real-agent calibration precedes held-out collection, but developers knew the task set; cross-fit, timing and transfer extensions are exploratory.

Targets and deployment utility. Operation targets are coupled to intended actions; localization identifies their metadata signature. The same-operation attribute diagnostic is inconclusive (Appendix C). Original agent reward, defended logical conformance and live prefix-shadow utility are distinct outcomes. Full framing/flush utility remains unmeasured. Local RPC timing excludes LLM generation and WAN transit. WF adapters support matched metadata comparisons; original-protocol equivalence and equal-utility deployment comparisons remain open.

External validity. The four mutable model aliases operate in one retail environment with separate user-simulator contexts. Luna has 29 training episodes per prompt; MiniMax and GLM each have 69 training tasks. Transfer matrices and live planning reuse known customers. Fresh customers, fixed weights, verified cross-provider identities, new environments and provider-backend measurements define the next validation axes.

10. Conclusion

Defense evidence transfers only across changes that preserve its protection and evaluation contract. Source-bound records and the audit make those conditions inspectable; experiments expose emitter, state, calibration and timing boundaries. Two-endpoint matrices turn model sensitivity into directional, target-specific tests. The practical decision procedure is to identify the changed boundary, retain matched inference, utility and cost objects, and re-measure that boundary before carrying a result forward.

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Appendix A. Audit Trail and Logical Equivalence

Evidence organization. The package separates the 82-family source map, 39-result ledger, 180-cell audit, transport tests, dispatch tests and real-agent studies. Each source record has a version and section/table/figure/paragraph locator. Authors reviewed all family labels and verified the result records and audit cells as described in §2.3. Initial AI-assisted extraction and subsequent author adjudication remain distinct archived columns; the supplementary process statistics and blind-recoding templates support follow-up audit. The supplement supplies detailed algorithms, full result grids and archived coding decisions; the main paper contains the comparison rules, complete family matrix and record index needed to follow the synthesis.

Logical equivalence. For deterministic tools, a real action gives $(s_{t+1}, o_t) = T(s_t, a_t)$. The defended dispatcher forwards that action unchanged, runs cover on a disjoint state s_t^{sh} and suppresses cover responses. With equal initial states and realized real actions, induction gives

$$s_t^D = s_t, \qquad \pi_{\mathrm{plan}}(D(H_t)) = H_t.$$

Here H_t is the logical message/action/response history, excluding wall-clock time. Payload-preserving framing and terminal shadow flush maintain the invariant, provided timeouts, shared-state side effects and timing-dependent tool behavior are absent. Checks compare every real-call projection and scoring field across all 690 trajectories and four program conditions. These establish implementation conformance to isolation. The utility boundary in §9.3 explains the further live-planner measurements.

The reproducibility index resolves each archived experiment in Appendix C to protocol, inputs, predictions and checksums. Saved-output verification reproduces the reported metrics without new model calls, while measured captures and analytic reconstructions retain their separate provenance.

Appendix B. Result-record index

Table 21 indexes all 33 retained baseline and six expansion records. Separate records identify changes in observation, intervention, target, execution population or cost object within a family. Shortened source locators below support direct lookup; the machine-readable ledger retains exact versions, hashes and full locators.

Appendix C. Experiment artifact index

Table 22 maps the four main experiments to stable artifact keys and detailed supplementary methods. Reanalysis, prompts and repetitions retain their original customer/task membership.

The reproducibility index resolves these keys to files, protocols, saved predictions and executable verification commands. The source versions and calibration chronology remain available for every analysis.

Extensions and sensitivity. The same-provider Luna extension contains 150 formal runs on 75 existing retail tasks, two prompts and one repeat. Standard-prompt write-only dispatch BA is 87.5/66.6%, versus 50.0/57.0% for get/find/list reads; rewards are 32/75 and 33/75, with seven protocol-error episodes retained. This tests the operation diagnostic under another model alias.

A customer-composition check on fixed cross-fit predictions changes standard-prompt joint-constant BA from 60.5/55.5% to 59.0/53.2% under equal-customer-within-class weighting. Rules-arm return changes from 59.4% to 60.4%. Across 192 leave-one-positive-customer checks,

TABLE 21. COMPLETE INDEX OF THE 39 RESULT RECORDS (26 FAMILIES). EACH ROW RETAINS ITS ARTIFACT ID, FAMILY CITATION, OBSERVATION/EXECUTION BOUNDARY AND SOURCE-LOCAL LOCATOR. THE PANELS FORM ONE INVENTORY, NOT INDEPENDENT DATASETS. T=TABLE, F=FIGURE, APP.=APPENDIX; LOCATORS REFER TO THE CITED SOURCE.

Record ID / family	Boundary, distinction; source locator
UI-delay-offline [5]	Offline delay; no live task outcome. *6.1 F5; A.5*
WRA-decoy-execution [6]	Executed decoys; usefulness proxy only. *8.1-8.3 T8-11*
NET-upstream-local [36]	Protected deployment/local serving; topology scoped. *IV-C-E TIII-IV*
NET-dummy-padding-simulation [36]	Synthetic insert/pad; no live execution. *VII TIX*
EARLY-K-calibration-oracle [3]	K-specific fit then oracle; not live KV trial. *V-A TVI; IV-A*
EARLY-semantic-deidentification [3]	GPTCache prototype cost; not KV-granularity cost. *V-B*
WHISPER-randompadding-capture [39]	New padded captures; rounded medians. *6.1 T3-4*
WHISPER-batching-replay [39]	Offline batching; model-dependent result. *6.2 F7*
WHISPER-injection-replay [39]	Synthetic packets; no live zero-delay claim. *6.3 T5*
SPLICE-CTBF-signal [46]	Prototype background load; tested signal only. *VII-C TII*
SPLICE-QCP-cost [46]	QCP component cost; excludes combined CTBF. *VII-C TII*
TIME-length-instruction [47]	Length instruction; unequal outputs remain. *7.2*
TIME-padding-estimate [47]	Analytic plaintext cost; not network bytes. *7.3*
SPEC-padding-REST [38]	Padding/aggregation; payload cost, placement unverified. *6.1 F6-7 T2*
DANGOL-capture [14]	Capture transform; distinct from live population. *9.8-9.9 T6*
DANGOL-livebinding [14]	Live loopback binding; do not join capture accuracy. *9.7, OHTTP paragraph after T 5*
SIG-common-guardrail [55]	Common instruction; not a causal guardrail effect. *4.1-4.2; App.C*
SAFE-post-trigger [15]	Post-trigger probes; detector cost not service TTFT. *7.2.4 F12; 7.3-7.4*
AHB-online-executed [12]	Executed UI edits; task/state changes possible. *5.1.2 T1; E.1-E.3*
AHB-offline [12]	Offline touch edits; no online task damage claim. *T1 offline rows; E.1-E.3*
KVGOV-partialflush-sim [33]	Deterministic simulation; not live protected recovery. *5.1-5.3*
KVGOV-hardware-gap [33]	Hardware gap calibration; not protected ASR. *5.7-5.8*
KVGOV-HMAC-cost [33]	HMAC microcost; not cache reuse/task loss. *5.9*
TRAFFIC-prompt-reformulation [51]	Collected prompt rewrites; task utility unmeasured. *V-C1 F9a*
TRAFFIC-temperature [51]	Collected decoding changes; no monotone benefit. *V-C1 F9a*
CARLINI-fixed-rate [35]	Rate/replayed cost; per-token, not request delay. *VII F11-12*
GAO-zero-copy-off [20]	Loader component costs; not end-to-end TTFT. *7.1*
TDXRAY-oblivious-map [27]	Lookup microbenchmark; not full inference cost. *6.1 F8 T3*
AGENTTELL-fixed-instruction [62]	Live local browser tasks; loaded/cold context contrast. *4.3.1; 5 Finding1*
FP-CLOUDFLARE [63]	Live access control; not user confidentiality. *6.4 T5*
BAR-co-running-noise [21]	MIG noise test; not an LLM defense trial. *7.2-7.3 T8*
RHYTHM-VPN [34]	Live VPN shift; baseline-trained attacker only. *VI-B; VII*
DEB-filter-origin [42]	Filter creates attack view; not mitigation of it. *Threatmodel; 4.2*
CACHEPRUNE-selective [48]	Selective cache sharing; direct token recovery/quality. *3.1; 5.1–5.6; T1; App.A*
CACHEPRUNE-context [48]	Contextual guessing; distinct from direct recovery. *3.1; 5.1–5.6; T1; App.A*
OTRO-unpadded [16]	ORAM TTFT/memory; access-count padding excluded. *VII-C–E; TIII*
OTRO-epochpad [16]	Epoch-padding estimate; not measured padded TTFT. *VII-C–E; TIII*
SPARLEAK-replacement [25]	Sparse replacement/quality; attack cost is separate. *3.1; 4–7; T4*
SHAPE-padding-estimate [26]	Constant-work proposal/estimate; no executed pair. *IX-B1*

standard-prompt cancellation ranges 57.3–62.0%; these are influence ranges, not confidence intervals.

The same-operation attribute check fixes single-item returns and a pre-fit digital-device grouping. Per prompt, 20 customer–order–item instances from 15 customers use the original customer folds and fixed LR/RF/GB learners. Maximum OOF BA is 30.3% [16.7,52.7], versus 50% for a constant predictor; a single fixed shuffled target gives 65.2%. Both prompts have identical features. Below-chance predictions can reflect reversed associations, and one shuffle is not a permutation distribution. This diagnostic is inconclusive for non-action attributes; its full predictions remain in the supplement.

Appendix D. Family-Level Holdout Evidence

Tables 23–25 cover all 82 indexed families. T: task/input/user; M: model; C: harness/topology; H: hardware; N: network/load; D: time. **H** records an explicitly reported held-out unit/domain, **V** a condition comparison or recalibration, **X** an explicit restriction or absent/unverified holdout in the inspected scope, and **?** an unresolved axis. A dash applies only to conceptual families. H does not certify all grouping details or successful transfer; V is not a holdout. Notes restrict each code. Core and adjacent families remain separate (C/A). The machine-readable matrix preserves the full source excerpts, locators and input hash; these tables synthesize that archived extraction rather than claim a new full-paper audit.

TABLE 22. FOUR MAIN-TEXT MODULES AND THEIR HISTORICAL ARTIFACT KEYS. THE RIGHT COLUMN SUPPLIES OPTIONAL IMPLEMENTATION AND FULL-GRID DETAIL.

Main module	Artifact keys	Observation/execution boundary and comparison	Detailed source
Exp. 1: transport emission	E1, E3, E5	Buffered two-link re-emission and token-stream emitters; upstream framing, final-link framing and matched attacker refitting are distinct comparisons.	S9/S11; S12
Exp. 2: execution and link	E2, E6, E7	Prescribed stateful plans and randomized dispatch cover, then agent–tool TLS composition; dispatch-visible calls and encrypted-link records are separate views.	S9/S11; S12
Exp. 3: real trajectories and localization	E9, E10, E12	Saved autonomous Sol trajectories; original pre-test calibration, exploratory customer cross-fitting and oracle operation-category localization retain their different fit/test boundaries.	S14; S15/S21
Exp. 4: time and WF	E13, E14	Local live tool-loop timing/full tunnels, then stronger learners and specified WF adaptations; measured clocks and simulations are distinguished, with matched suites for joint comparisons.	S19; S20
Extensions	E8	Stand-in-planner pipeline check.	S13
	E11	Retrospective training-fold operating-point selection with the historical suite.	S15
	E15	Additional same-provider Luna cohort.	S22
	E16	Same-operation semantic-attribute diagnostic on reused Sol data.	S23.8
	R11/R12	Two served model endpoints: native trajectories, frozen training calibration and TLS write localization.	S24
	R14	Attacker/calibration transfer matrices, non-oracle write detection and write-only framing with the expanded seven-attack suite.	S25
	R15	Fresh paired live-planner episodes under isolated prefix cover and injected tool-response deadlines.	S26

TABLE 23. HOLDOUT EVIDENCE, FAMILIES 1–28. CODES AND AXES ARE DEFINED IN APPENDIX D.

Family	Scope	T	M	C	H	N	D	Source locator and qualification
F01 AgentPrint [4]	C	?	?	?	?	?	?	Random trace splits; prompt grouping unresolved. III; V; Appendix D.A–D.D; Fig. 11
F02 UI Traces [5]	C	V	V	V	?	?	?	Cross-task/site comparisons; exact-question overlap remains. 3–4; source index
F03 WRA [6]	C	H	?	?	?	?	?	Prompt tests and synthetic persona split; version reconciliation open. 3.1; 6.1; 8–9; Tables 1/8/10/11; Appendix C.1–C.3
F04 FP-Agent [63]	C	H	?	X	?	?	?	Separate task holdout; visitor/host separation unresolved. 4.1–4.7; Tables 2/4; 6.1–6.2; Appendix E/Table 8
F05 Prompt length [1]	C	H	?	V	?	V	V	OOD inputs; browser/API and day/night buffering controls. 5.1–5.3; Fig. 4; Tables 2–3
F06 PromptPeek [2]	C	V	?	?	?	V	?	Candidate-prior and simulated request-load sweeps. III.B; VI.A–VI.C; Table II
F07 NetEcho [36]	C	V	?	V	?	?	?	Deployment/frontend scenarios; independent split binding unresolved. III.A; Table III; VI.A–VI.B/Table VI
F08 Early Bird [3]	C	?	V	?	?	?	?	Provider observations and candidate calibration; grouped split unresolved. arXiv 2409.20002: II.C; IV.A; V.A; Tables IV–V
F09 InputSnatch [44]	C	H	?	?	?	?	?	Held-out and generated target prompts; candidate-prior dependence. arXiv 2411.18191: 3.2; 4.3–4.4; Tables 2/4
F10 Whisper Leak [39]	C	H	V	?	?	?	?	Unique target-prompt holdout; model/provider comparisons. 3.1; 3 classifier/split discussion; 4.1–4.2; Table 1; 6
F11 Spill the Beans [23]	C	V	?	?	?	?	?	Exposure/monitor sweeps; independent capture grouping unresolved. Threat model; 5/Table 1; 6.1
F12 SpliceLeak [46]	C	V	?	?	?	?	?	Bounded cloze tasks and chunk/candidate coverage. Threat model; VII.A–VII.C; Figs. 7–8
F13 Time Will Tell [47]	C	?	?	?	?	V	V	Concurrent network/load profiling; short-window recalibration. 3; 4.1–4.4; 6.2–6.5; Tables 3–4
F14 Prompt Echo [37]	C	H	V	?	?	?	?	Overlap-filtered prompts; models evaluated separately. 3.1; 5.1–5.3; Tables 2–3
F15 I Know What You Said [20]	C	H	V	V	V	?	?	Prompt overlap filtering; model/hardware/framework comparisons. Abstract; source index
F16 Speculation [38]	C	H	?	?	?	V	?	New traces and OOD paraphrases; remote/load conditions. v4: 4.3–4.8; Table 1; 6.1
F17 Behind Bars [21]	C	?	V	?	V	?	?	Model identification CV; additional hardware evaluation. Table1; 6.1–6.3; Table7 (printed5383/PDF12); 7.1/7.3; Open Science
F18 Prompt caching [43]	C	?	V	?	?	?	V	Provider survey over two months; historical exposure only. 3.1–3.2; 4.1–4.2; Tables 1–2
F19 Whose Agent [61]	C	?	?	?	?	?	?	Trial split; task/site/version independence unresolved. 4,6.1–6.3; Tables4–5; Fig7;7
F20 Workflow traffic [14]	C	?	?	H	?	?	?	Leave-composition-out; stored and live populations separated. 3,9.1,9.7–9.14;Table6
F21 GPU contention [40]	C	?	?	V	?	V	?	Serving-stack and interleaved-load controls. IV,IX,TableVII,X,XIV–XVII
F22 AgentTell [62]	C	V	?	?	?	V	?	Secret rotation; cold/loaded and equal-completability controls. 3–5;AppendixD.2–D.3,Table8;AppendixE
F23 Traffic benchmark [51]	C	?	?	H	?	?	?	Topology shift distinct from feature-view restriction. III,IV-A–C,V-A TablesI–III,V-B Fig8,VI
F24 Remote timing [35]	C	?	V	?	?	?	V	Month-long observations; method/model changes confounded. III,IV-B–C,V-D–E,VI-C,VII,Figs8,10,12
F25 Persona fingerprint [60]	C	?	?	X	?	?	?	Pooled sites; leave-site-out not verified. 2,3.1–3.4,4,5.1,5.5–5.8
F26 TDXRay [27]	C	?	?	?	?	?	?	Tokenizer-localized LLM case; AES result separate. 3;5.1–5.4;6.1–6.2
F27 Detokenization [29]	C	?	?	X	?	?	?	Explicit same-instance profiling restriction. 3.1 and 3.3: shared-code, SMT and same-instance profiling; source index
F28 Spillage [80]	A	V	?	?	?	?	?	Task-irrelevance and selected-site evaluation. 4.1–4.2 (evaluator inputs), 5.4, Table 4, 6.2

TABLE 24. HOLDOUT EVIDENCE, FAMILIES 29–56. CODES AND AXES ARE DEFINED IN APPENDIX D.

Family	Scope	T	M	C	H	N	D	Source locator and qualification
F29 PlanTwin [81]	A	V	?	?	?	?	?	Template candidate pool; production transfer unresolved. 2.2,3.2,4.2–4.4,5.1–5.9,6.1
F30 SICC [82]	A	?	?	?	?	?	?	Authorized-view theorem; empirical shift binding unresolved. Theorem 1; Experiments Setup; Tables 3–4, 7–8
F31 TOPR [83]	A	V	?	?	?	?	?	Benchmark and alignment populations differ. 3.1–3.2,4,5.2,5.6,6,Limitations
F32 ClawGuard [13]	C	?	?	V	?	?	V	Carrier recollection and cycle drift; not zero-shot transfer. III-C,IV,VI,AppendixA
F33 SIGLEAK [55]	C	V	?	?	?	?	?	Paired prompts; not physical-skill removal. Definition3,3.2,4.1–4.2
F34 ProcGrep [56]	C	H	?	V	?	?	?	Task-grouped attribution under scaffold regimes. 4–5,Tables7–8
F35 SecEmb [84]	A	?	X	?	?	?	?	Single GPT-2 workload; DLRM result separate. III,V,VI-D;Figs14–15
F36 Mobile agents [85]	A	?	?	?	?	?	?	Direct leakage cases; holdout binding unresolved. IV-D3 log leakage; V-D2 deep-link leakage; V-D3 log leakage; Table II; VI-A
F37 Hidden risks [86]	A	?	?	?	?	?	?	Content exfiltration/access-control cases. 3.2 framework and risk scope; source index
F38 Browser privacy [87]	A	?	?	V	?	?	?	Browser/profile-state measurements; mixed leakage mechanisms. V-C1–3 browser/network/profile-state measurements; source index
F39 LLMs Have Rhythm [34]	C	?	V	?	?	H	V	New captures and cross-network tests; overlapping windows caveat. III-IV; V-C–G; VI-A/B; Table I; Figs10–16; VII
F40 MoEcho [19]	C	H	V	?	?	V	?	Cross-template/dataset tests; contention/context conditions. 3.1;4;6.2–6.6;Tables3,5–10;7.2;8.1–8.2
F41 LLM-Fin [73]	A	?	H	?	?	?	?	Unseen variants within known architecture families. III; IV-B/C; Tables II–III; IV-C/D; V-A; V-A versus Introduction/Fig.2; Table IV
F42 ML privacy channels [42]	C	?	V	?	?	?	?	Enumerative/filter probes across named models; no learned split. 4.1; 4.2.1; 4.2.2/Fig.7; 4.2.3; 4.2.3/Table2
F43 MoE stealing [49]	C	?	?	?	?	?	?	Constructive word extraction in controlled shared batches. 2.2; 4; 4; 4/Table2; 5/Fig.7
F44 Energon [18]	C	?	V	?	?	?	?	Known-candidate CV; exploratory hardware not transfer proof. III-A/B; IV-A/C; IV-B/Tables I,III; V; VII
F45 SafeKV [15]	C	?	V	?	?	?	?	Named serving models; calibration/test separation unreported. 3; 7.1; 7.2.1;7.2.3/Fig.12; 7.3/Table7; 7.4; 4.2.2 versus6/7.2.3
F46 InferNet [88]	A	?	V	?	V	?	?	Same-label mixed-model and platform comparisons. 5.2; source index
F47 ArchSnoop [24]	C	?	H	?	V	?	?	Random-weight training to official/fine-tuned models; power modes. 3.2;6.1; 6.1; Fig9, paperp58, visually checked; 6.2.3;6.3.1; 4.1;7.1-7.2
F48 Coding fingerprints [7]	A	X	?	X	?	?	X	PR CV; repository/time/human-mediated holdouts unreported. 2.1-2.4; Tables 1-2; 4; 6; replication reference
F49 AHB [12]	C	?	?	X	?	?	?	Offline/live humanization; website-transfer binding unresolved. 2.1-2.2; 3.1; 4.1-4.3; 5.1.2; Table 1 Trip Planning; E.1-E.3
F50 WildAuth [8]	A	H	?	X	?	?	?	User-disjoint cross-topic split; no verified cross-service test. 2.3; 4.1-4.3; 5.3-5.5; 6.1-6.2; Tables 1-3; Appendix D/Table4
F51 Cycles [54]	C	X	?	X	?	?	X	One application; no held-out split located. 1; 2; 2.1 CDCS/CDDAG versus CDSA/Hybrid; 3.1; Tables1-2; 3.2; 4
F52 LumiMAS [53]	C	?	V	V	?	?	?	Nonoverlap logs; backend/platform comparisons; task grouping open. 3;4.1-4.3;5.1.1-5.1.7;5.2;Table1;6.1-6.5;B.1/Table3;B.2/Table4;C.1.1
F53 SymDiag [89]	A	V	V	?	?	?	?	Domains/reasoners vary with inputs and feedback. 3.1-3.4;4.1.1-4.1.5;5.1;6.1-6.4;7
F54 LeakyLMs [45]	C	?	H	?	V	?	?	Held-out architecture configurations; known-GPU calibration. 3;4.4-4.5 Tables1-2;5.4-5.5 Tables3-6/Fig6;6;A.1;B.4-B.7
F55 KVGov [33]	C	?	?	?	V	?	?	Simulation, A100 timing and HMAC microcost separate. 2.2-2.3;4;5.1-5.3 Tables3-4;5.7-5.9 Tables8-9;7
F56 Data-centric survey [90]	A	-	-	-	-	-	-	Conceptual synthesis; empirical holdout not applicable. 1 Fig1;4-6;8-9 discussion/conclusions

TABLE 25. HOLDOUT EVIDENCE, FAMILIES 57–82. CODES AND AXES ARE DEFINED IN APPENDIX D.

Family	Scope	T	M	C	H	N	D	Source locator and qualification
F57 Serving RCA [31]	C	?	?	X	?	?	?	Single serving deployment with injected faults. Sections3,4.1-4.3,5.1-5.2; Table1
F58 ZeroPrint [9]	A	?	V	?	?	?	?	Known-family provenance pairs; partition not re-extracted. Header;4.1-4.4;Figure1;5.1-5.3
F59 Endpoint fingerprints [10]	A	?	?	V	?	?	V	Controlled endpoint changes over matched time windows. Sections3,4.1-4.3,6.1-6.2/Table1,7-9
F60 StepFinder [11]	A	H	?	?	?	?	?	Task-disjoint subsets; denominator reconciliation unresolved. Header;Sections4,5.1-5.4,6;Tables4-5
F61 Web-agent detection [52]	C	?	V	?	?	?	?	Six agents with human/scraper controls. 3.1.4;3.3;6;7
F62 Cross-campaign [59]	C	V	?	V	?	?	?	Crossed-style and regenerated conditions; execution stress subset. AppendixB decomposition; source index
F63 AgentProv [57]	C	?	V	?	?	?	?	Substituted checkpoints; system-prompt perturbation. 3.1-3.4;4.1-4.4;A.5
F64 LiDAR [58]	C	?	V	V	?	?	?	Enrollment-only identity; two harnesses and perturbations. 2;5.1-5.3;6.1-6.5
F65 LLM-WFIN [30]	C	V	?	?	?	?	?	Site traces; loading versus query phases. 3.2-3.3;5.1-5.2;6.1-6.5;Tables2-5;publisher rendered primary article
F66 MIRAGE [50]	C	V	V	?	?	?	?	Encodings/models/scenarios; matched decoder-layer controls. 3.4;4.1-4.2;7.1-7.4;AppendixL.3-L.4
F67 Routing metadata [41]	C	V	?	V	?	?	?	Request domains and routers; exploratory sensitive-label gate. III;IV;V;VI;VII
F68 Shape and Substance [26]	C	V	V	V	V	?	?	Geometry/model/cache/hardware conditions; padding estimate. IV;V;VII;VIII;IX
F69 SplitDecoy [32]	C	?	?	?	?	?	?	Repeated seeds; forward/backward boundary, not a holdout. 2;5.1-5.3;Table7-8;A.2.2
F70 PAGE404 [22]	C	V	V	?	?	?	?	Architecture and prompt-length targets; transfer binding open. II-B;IV-D4-D6;V
F71 Edge EM [17]	C	H	?	?	?	?	?	Unseen-word test separate from profiled vocabulary recovery. IV;V;VI
F72 SparLeak [25]	C	V	V	V	?	?	?	Models/datasets/sparse mechanisms; per-phase outcomes. 3.1;4-7;Table4
F73 Sparseety [28]	C	V	V	V	?	?	?	Models/datasets/implementations; oracle-conditioned extraction. III;V;VI;VIII;IX
F74 OTRO [16]	C	?	?	?	?	?	?	Closed-corpus conditional leakage; epoch cost separately derived. III;V;VI;VII-C-E;TableIII
F75 CachePrune [48]	C	V	V	?	?	?	?	Models/workloads and detector-miss stress; separate fits. 3.1;5.1-5.6;Table1;AppendixA
F76 Privacy foundations [66]	A	-	-	-	-	-	-	Conceptual contract; cross-institution example illustrative. III;IV/TableIII;IV-A;VI
F77 DP-MemView [91]	A	?	?	?	?	?	?	Semantic response privacy; holdout binding not re-extracted. Problem Definition;DP-MemView;Experiments;AppendixB.5
F78 PriCon [92]	A	?	?	?	?	?	?	Semantic cloud consultation; holdout binding not re-extracted. PDF3;5.1-5.2;Tables2-4
F79 SCAAR [93]	A	?	?	?	?	?	?	LLM EM/length test; other classifier results separate. 7;LLM extension;TVLA experiment
F80 CacheSolidarity [94]	A	?	?	?	?	?	?	Candidate-prefix timing gap; no scored inference error. v1 title;5.2.3;Figure7
F81 Vision encoder [95]	A	?	?	?	?	?	?	Representation inversion; holdout binding not re-extracted. 2;3;5;7;8/Table5
F82 Split gradients [96]	A	?	?	?	?	?	?	Split-compute content inversion; holdout binding unresolved. 3;4.1-4.5;5;AppendixB